CHAPTER 1 | Introduction to Information Systems

CHAPTER ROADMAP

Information systems connect people, processes, data and technologies to create organisational value.

Learning outcomes

1. Explain and apply: From data to decisions.

2. Explain and apply: Components of an information system.

3. Explain and apply: Systems by organisational level.

4. Explain and apply: Information systems and business functions.

5. Explain and apply: Organisational impact.

6. Explain and apply: Evaluation and responsible use.

Concept map

BUSINESS QUESTION → DATA / PEOPLE / PROCESS → INFORMATION SYSTEM → ACTION → MEASURABLE OUTCOME

How to use this chapter

Read the six topic pages, explain the definitions aloud, study the scenario, and answer the questions without looking. Then check the model answers. This guide is original educational commentary, not a reproduction of the textbook.

Big picture

Information systems connect people, processes, data and technologies to create organisational value. The central skill is to connect terminology to a real organisational decision. Whenever you meet a new concept, identify its purpose, required inputs, expected benefits, limitations, and the people responsible for it.

From data to decisions

Core explanation

Data are raw observations; information gives observations context; knowledge combines information with experience; decisions turn knowledge into action. Quality depends on accuracy, timeliness, relevance, completeness and accessibility.

Deeper understanding

To apply from data to decisions, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• data: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• information: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• knowledge: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• information quality: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A shop turns checkout records into a weekly sales report and then changes stock orders.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of from data to decisions in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Components of an information system

Core explanation

An information system includes hardware, software, data, networks, procedures and people. Technology alone does not deliver value: roles, incentives and workflows must also fit.

Deeper understanding

To apply components of an information system, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• hardware: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• software: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• procedures: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• people: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• sociotechnical system: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A hospital scheduling tool fails when staff workflows and training are ignored.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of components of an information system in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Systems by organisational level

Core explanation

Transaction processing supports daily work; management reporting supports supervision; decision support explores alternatives; executive tools support strategic monitoring. Systems increasingly integrate across levels.

Deeper understanding

To apply systems by organisational level, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• TPS: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• MIS: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• DSS: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• executive information: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A retail chain links point-of-sale data to regional performance dashboards.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of systems by organisational level in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Information systems and business functions

Core explanation

Finance, marketing, operations, HR and customer service all use specialised applications. Integrated platforms reduce duplicated records but require common definitions and governance.

Deeper understanding

To apply information systems and business functions, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• functional information system: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• integration: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• ERP: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

An employee change is entered once and shared with payroll and scheduling.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of information systems and business functions in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Organisational impact

Core explanation

Digital systems may automate tasks, inform employees, redesign processes or enable new products. Benefits can include speed and reach, while risks include dependency, surveillance and unequal access.

Deeper understanding

To apply organisational impact, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• automation: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• digital transformation: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• digital divide: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Online appointment booking improves access but needs alternatives for excluded users.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of organisational impact in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Evaluation and responsible use

Core explanation

Evaluate an IS through effectiveness, efficiency, user adoption, security, cost and social impact. Identify stakeholders and trade-offs rather than treating technology as automatically beneficial.

Deeper understanding

To apply evaluation and responsible use, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• effectiveness: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• efficiency: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• stakeholder: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• total cost: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A university measures not just portal uptime but whether students complete enrolment.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of evaluation and responsible use in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

CHAPTER 2 | Organizational Strategy, Competitive Advantage, and Information Systems

CHAPTER ROADMAP

IT creates strategic advantage when it supports distinctive capabilities and better business processes.

Learning outcomes

1. Explain and apply: Business processes.

2. Explain and apply: BPI, BPR and BPM.

3. Explain and apply: Business pressures.

4. Explain and apply: Porter five forces.

5. Explain and apply: Value chain and strategies.

6. Explain and apply: Sustaining advantage.

Concept map

BUSINESS QUESTION → DATA / PEOPLE / PROCESS → INFORMATION SYSTEM → ACTION → MEASURABLE OUTCOME

How to use this chapter

Read the six topic pages, explain the definitions aloud, study the scenario, and answer the questions without looking. Then check the model answers. This guide is original educational commentary, not a reproduction of the textbook.

Big picture

IT creates strategic advantage when it supports distinctive capabilities and better business processes. The central skill is to connect terminology to a real organisational decision. Whenever you meet a new concept, identify its purpose, required inputs, expected benefits, limitations, and the people responsible for it.

Business processes

Core explanation

A process has inputs, activities, owners, controls and outputs. Cross-functional processes often expose handoff delays and inconsistent data. Map the actual workflow before proposing automation.

Deeper understanding

To apply business processes, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• business process: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• workflow: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• process owner: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Order-to-cash spans sales, inventory, delivery and finance.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of business processes in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

BPI, BPR and BPM

Core explanation

Business process improvement makes incremental changes; reengineering redesigns a process radically; BPM continuously models, measures and improves processes. Each requires different levels of change and risk.

Deeper understanding

To apply bpi, bpr and bpm, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• BPI: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• BPR: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• BPM: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• process mapping: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A company removes one approval step (BPI) or redesigns fulfilment end-to-end (BPR).

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of bpi, bpr and bpm in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Business pressures

Core explanation

Competition, globalisation, changing customer expectations, regulation and technological change create pressure. Organisations respond with customer focus, alliances, innovation and improved operations.

Deeper understanding

To apply business pressures, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• business pressure: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• organisational response: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A local retailer adds click-and-collect in response to online competition.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of business pressures in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Porter five forces

Core explanation

Analyse rivalry, new entrants, substitutes, supplier power and buyer power. IT can alter each force, but competitors can copy technology, so complementary capabilities matter.

Deeper understanding

To apply porter five forces, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• five forces: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• switching cost: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• entry barrier: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A platform raises switching costs through integrations and customer history.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of porter five forces in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Value chain and strategies

Core explanation

Primary and support activities create value. IT can lower cost, differentiate products, improve focus or strengthen relationships. Assess where information flows create bottlenecks.

Deeper understanding

To apply value chain and strategies, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• value chain: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• cost leadership: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• differentiation: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Real-time inventory visibility reduces stockouts and supports faster delivery.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of value chain and strategies in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Sustaining advantage

Core explanation

Alignment, data quality, culture, skills and continuous improvement determine whether an advantage lasts. Consider strategic risks: vendor lock-in, security and overinvestment.

Deeper understanding

To apply sustaining advantage, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• strategic alignment: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• capability: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• lock-in: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A firm gains little from an expensive CRM if sales teams do not use it.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of sustaining advantage in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

CHAPTER 3 | Ethics and Privacy

CHAPTER ROADMAP

Responsible information use requires balancing organisational benefits with individual rights and social consequences.

Learning outcomes

1. Explain and apply: Ethical reasoning.
2. Explain and apply: Privacy and personal data.
3. Explain and apply: Privacy principles.
4. Explain and apply: Intellectual property.
5. Explain and apply: Digital ethics and inequality.
6. Explain and apply: Governance and accountability.

Concept map

BUSINESS QUESTION → DATA / PEOPLE / PROCESS → INFORMATION SYSTEM → ACTION → MEASURABLE OUTCOME

How to use this chapter

Read the six topic pages, explain the definitions aloud, study the scenario, and answer the questions without looking. Then check the model answers. This guide is original educational commentary, not a reproduction of the textbook.

Big picture

Responsible information use requires balancing organisational benefits with individual rights and social consequences. The central skill is to connect terminology to a real organisational decision. Whenever you meet a new concept, identify its purpose, required inputs, expected benefits, limitations, and the people responsible for it.

Ethical reasoning

Core explanation

Ethics concerns what should be done, not only what law permits. Identify stakeholders, likely harms, alternatives, duties and fairness before deciding.

Deeper understanding

To apply ethical reasoning, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• ethics: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• stakeholder: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• ethical dilemma: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A firm debates whether detailed employee monitoring is proportionate.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of ethical reasoning in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Privacy and personal data

Core explanation

Privacy includes reasonable control over personal information. Personal data can identify a person directly or indirectly; processing includes collecting, storing, sharing and deleting it.

Deeper understanding

To apply privacy and personal data, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• privacy: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• personal data: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• processing: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Location histories may reveal sensitive patterns even without names.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of privacy and personal data in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Privacy principles

Core explanation

Purpose limitation, minimisation, transparency, security, accuracy and retention limits help govern personal data. In the EU, GDPR also requires a lawful basis and protects data-subject rights.

Deeper understanding

To apply privacy principles, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• data minimisation: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• lawful basis: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• retention: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• GDPR: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A study app collects only information needed to provide its service.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of privacy principles in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Intellectual property

Core explanation

Copyright protects original expression; patents protect qualifying inventions; trademarks distinguish commercial sources; licences specify permitted uses. Open source is not the same as public domain.

Deeper understanding

To apply intellectual property, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• copyright: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• patent: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• trademark: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• licence: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A developer checks licence obligations before reusing a software library.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of intellectual property in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Digital ethics and inequality

Core explanation

Algorithmic bias, misinformation, accessibility barriers and unequal connectivity create social consequences. Assess who benefits, who bears risks and who may be excluded.

Deeper understanding

To apply digital ethics and inequality, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• algorithmic bias: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• accessibility: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• digital divide: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

An automated hiring system is audited for discriminatory outcomes.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of digital ethics and inequality in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Governance and accountability

Core explanation

Privacy impact assessments, training, access controls, reporting channels and clear ownership make ethical principles operational. Document decisions and provide ways to challenge outcomes.

Deeper understanding

To apply governance and accountability, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• accountability: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• DPIA: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• governance: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A university documents why it collects learning analytics and who can view them.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of governance and accountability in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

CHAPTER 4 | Information Security

CHAPTER ROADMAP

Security protects confidentiality, integrity and availability through people, processes and technology.

Learning outcomes

1. Explain and apply: Security goals and risk.
2. Explain and apply: Threats and attacks.
3. Explain and apply: Identity and access.
4. Explain and apply: Protective technologies.
5. Explain and apply: Continuity and recovery.
6. Explain and apply: Security culture and governance.

Concept map

BUSINESS QUESTION → DATA / PEOPLE / PROCESS → INFORMATION SYSTEM → ACTION → MEASURABLE OUTCOME

How to use this chapter

Read the six topic pages, explain the definitions aloud, study the scenario, and answer the questions without looking. Then check the model answers. This guide is original educational commentary, not a reproduction of the textbook.

Big picture

Security protects confidentiality, integrity and availability through people, processes and technology. The central skill is to connect terminology to a real organisational decision. Whenever you meet a new concept, identify its purpose, required inputs, expected benefits, limitations, and the people responsible for it.

Security goals and risk

Core explanation

The CIA triad defines protection objectives. Risk combines threats, vulnerabilities, likelihood and impact; controls reduce but rarely eliminate risk.

Deeper understanding

To apply security goals and risk, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• confidentiality: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• integrity: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• availability: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• risk: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

An outage affects availability while altered grades threaten integrity.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of security goals and risk in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Threats and attacks

Core explanation

Phishing, malware, ransomware, credential theft, insider misuse and denial-of-service attacks exploit technical and human weaknesses. Distinguish a threat from a vulnerability.

Deeper understanding

To apply threats and attacks, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• phishing: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• malware: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• ransomware: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• vulnerability: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

An employee follows a fake sign-in link and exposes a password.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of threats and attacks in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Identity and access

Core explanation

Authentication checks identity; authorisation determines permissions. MFA, least privilege and periodic reviews limit account misuse. Avoid sharing credentials.

Deeper understanding

To apply identity and access, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• authentication: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• authorisation: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• MFA: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• least privilege: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A teaching assistant can view grades but cannot change tuition records.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of identity and access in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Protective technologies

Core explanation

Encryption protects data confidentiality; firewalls filter traffic; endpoint protection detects suspicious behaviour; patching closes known vulnerabilities. Each control has limitations.

Deeper understanding

To apply protective technologies, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• encryption: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• firewall: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• patch management: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Encrypted laptops protect files if a device is stolen.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of protective technologies in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Continuity and recovery

Core explanation

Backups, incident response, disaster recovery and business continuity prepare for failures. Recovery time and recovery point objectives express acceptable downtime and data loss.

Deeper understanding

To apply continuity and recovery, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• backup: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• incident response: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• RTO: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• RPO: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A clinic restores appointments from backups after ransomware.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of continuity and recovery in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Security culture and governance

Core explanation

Policies, training, audits, supplier reviews and risk ownership matter as much as tools. Measure security with tested controls and realistic exercises.

Deeper understanding

To apply security culture and governance, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• security policy: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• audit: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• defence in depth: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A company rehearses an incident and discovers outdated contact lists.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of security culture and governance in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Sources and scope

Rainer, R. Kelly and Brad Prince (2022), Introduction to Information Systems, 9th ed., International Adaptation. Chapter titles verified against the Wiley publisher table of contents. Original study notes; supplement with the UOC assigned readings and official chapter case studies.